

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection Behavioral and visual stimulus data were recorded as voltages on a Digidata 1440 (Molecular Devices) I/O board. Two-photon imaging data were collected using PrairieView 5.4 (Bruker). Floating ball positions were measured using Fictrac v1. Wing tracking was conducted with Strokelitude (<https://github.com/motmot/strokelitude>).

Data analysis Two photon imaging data were pre-processed using Fiji (to define regions of interest, version: 2.1.0/1.53c). All data were analyzed with custom code in python 2.7 and 3.6. Analysis code is deposited on github (https://github.com/Cheng-Lyu/TravelingDirectionPaper_code).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Data for main Figures are available on Dropbox (<https://www.dropbox.com/sh/p8bqwavlsyI9ppv/AABz2-vda4Q3gukXqp8Ba2Gwa?dl=0>). Other data are available upon request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical tests were used to determine sample size. We used sample sizes (~4-13 flies per condition) that been previously shown to have sufficient statistical power in similar experiments in the past (e.g., Seelig & Jayaraman 2015, Green et al. 2017).
Data exclusions	We did not exclude flies or data from any analysis, unless flies appeared unhealthy at the time of the experiment or were unable to last through the required experimental trials, i.e., would not maintain continuous flight for > 5 s bouts, probably due to 5-6 transgenes expression that can affect overall health and flight vigor.
Replication	All experiments discussed in the paper were conducted once at the conditions shown; no experimental replicate was excluded. For most experiments, data across multiple days were collected and the data across days were consistent. In immunohistochemistry plots (Extended Data Fig. 1a-c), two brains were imaged, but only one is shown. Both imaged brains showed the same qualitative pattern of staining
Randomization	Organisms are not allocated to control and experimental groups by the experimenter in this work, rather the flies' genotype determines their group. Thus, randomization of individuals into treatments groups is not relevant.
Blinding	The investigators were not blind to the flies' genotypes. All data collection and analysis was done computationally.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	For primary antibodies, we used mouse anti-Brp(nc82, DSHB) and chicken anti-GFP (Rockland, 600-901-215). For secondary antibodies, we used Alexa Fluor 488 goat anti-chicken (A11039, Invitrogen) and Alexa Fluor 633 goat anti-mouse (A21052, Invitrogen).
Validation	All antibodies used in this study were validated as described at the following websites (and references therein): DSHB: https://dshb.biology.uiowa.edu/nc82?quantity=1&product-form=1 , Rockland: https://rockland-inc.com/store/Antibodies-to-GFP-and-Antibodies-to-RFP-600-901-215-O4L_23908.aspx , Invitrogen: https://www.thermofisher.com/antibody/product/Goat-anti-Chicken-IgY-H-L-Secondary-Antibody-Polyclonal/A-11039 and https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21052

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	We used female Drosophila melanogaster that were 2-6 days old. All fly strains and fly genotypes are described in details in the Methods.
Wild animals	The study did not involve wild animals.

Field-collected samples	The study did not involve samples collected from the field.
Ethics oversight	No ethical oversight was required because no vertebrates were used.

Note that full information on the approval of the study protocol must also be provided in the manuscript.